Calibrating Conditional Risk

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Abstract

We introduce and study the problem of calibrating conditional risk, which involves estimating the expected loss of a prediction model conditional on input features. We analyze this problem in both classification and regression settings and show that it is fundamentally equivalent to a standard regression task. For classification settings, we further establish a connection between conditional risk calibration and individual/conditional probability calibration, and develop theoretical insights for the performance metric. This reveals that while conditional risk calibration is related to existing uncertainty quantification problems, it remains a distinct and standalone machine learning problem. Empirically, we validate our theoretical findings and demonstrate the practical implications of conditional risk calibration in the learning to defer (L2D) framework. Our systematic experiments provide both qualitative and quantitative assessments, offering guidance for future research in uncertainty-aware decision-making.

1 INTRODUCTION

Understanding and quantifying the uncertainty of machine learning models is essential in real-world decision-making systems, especially in applications involving risk-sensitive domains such as autonomous systems and human-AI interaction (Abdar et al., 2021; Smith, 2024). Decision-making systems must not only provide accurate predictions but also assess the reliability of those predictions to inform downstream tasks. Traditional uncertainty quantification methods focus on global or marginal uncertainty estimates. However, many practical applications require a more refined characterization of uncertainty that is specific to individual inputs (Yang et al., 2015; Cortes et al., 2016; Madras et al., 2018). This motivates the study of conditional risk calibration, where the goal is to estimate the expected loss of a given predictor conditioned on input features. A well-calibrated conditional risk estimator provides an interpretable measure of predictive quality, enabling more robust decision-making in safety-critical and resource-sensitive applications.

Despite its importance, the problem of conditional risk calibration/estimation has received limited formal treatment in the literature. Existing methods that implicitly rely on conditional risk estimation often approach it heuristically (Shrivastava et al., 2016; Yoo and Kweon, 2019), without a rigorous formulation and understanding of its theoretical properties.

This work provides a comprehensive theoretical and empirical investigation of conditional risk calibration. Our key contributions are as follows:

- We present a systematic formulation of the conditional risk calibration problem, establishing it as an important question in machine learning.
- We analyze conditional risk calibration in both classification and regression settings and demonstrate that the problem can be naturally formulated as a standard regression task. Additionally, for classification settings, we establish a connection between conditional risk calibration and individual probability calibration, showing that improving probability calibration enhances conditional risk estimates.
- We provide theoretical and empirical insights on the performance of conditional risk estimators and analyze conditions under which these estimators perform well. Our empirical studies validate these theoretical insights and demonstrate their practical relevance in the downstream application of learning to defer (L2D), providing new benchmarks and highlighting the practical benefits of

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conditional risk calibration in designing more reliable, interpretable, and risk-aware AI systems.

We defer the related work section to Appendix A.

2 FORMULATION

Let $\mathcal{X}$ be the space of input, $\mathcal{Y}$ the space of output. We denote $P_{\mathcal{X}}$, $P_{\mathcal{X}\times\mathcal{Y}}$, $P_{\mathcal{Y}|x}$ as the marginal, joint, and conditional distribution. With a given predictor $\hat{f}(\cdot) : \mathcal{X} \to \mathcal{Y}$ and loss function $\ell(\cdot,\cdot) : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$, we define the conditional loss/risk by

$$g(x) = \mathbb{E}[\ell(\hat{f}(X), Y)|X = x]. \tag{1}$$

With an appropriate $\ell(\cdot)$, $g(x)$ can be interpreted as a measure of the "uncertainty" or "quality" of the predictor $\hat{f}(\cdot)$.

In practice, the conditional loss function $g(x)$ is not directly observable, and its estimation is inherently noisy. This challenge is exacerbated in modern AI applications, where inputs often take the form of images or natural language, and for a specific input x, we can only observe one corresponding sample $g(x)$.

However, $g(x)$ can be a useful metric in settings like human-AI interactions. For instance, a large $g(x)$ may indicate that $\hat{f}$ is uncertain about the input x, and this information is valuable in several contexts, as discussed below.

Uncertainty-based Active Learning. Active learning (Joshi et al., 2009; Yang and Loog, 2016; Nguyen et al., 2022) aims to improve model performance while minimizing labeling costs by selectively querying the most informative samples. Conditional risk calibration/estimation provides a principled approach to sample selection by identifying inputs where the predictor $\hat{f}(\cdot)$ exhibits high uncertainty, quantified by the *estimated* conditional loss $\hat{g}(x)$ on $g(x)$. Data points with high estimated conditional loss $\hat{g}(x)$ are prime candidates for labeling, as they indicate regions where the model is uncertain or prone to error. Acquiring the corresponding label y and retraining on the augmented dataset (x, y) are more likely to improve $\hat{f}$, particularly under a limited labeling budget. By prioritizing these samples, active learning reduces annotation costs while maintaining competitive performance.

Learning to Defer. Learning to defer (L2D) (Cortes et al., 2016; Madras et al., 2018) allows an AI system to decide whether to make a prediction or defer to a human expert based on estimated confidence. Conditional risk calibration/estimation provides a natural decision criterion by estimating the expected loss of the model's prediction. If $g(x)$ exceeds a threshold, the system defers to an expert, ensuring higher reliability while minimizing unnecessary interventions. This approach optimally allocates expert resources, reducing deferrals while maintaining high predictive accuracy. It is particularly useful in high-stakes applications.

2.1 Learning Conditional Loss

We now introduce conditional risk calibration/estimation for classification and regression settings.

Classification. For classification problems, we take $\mathcal{Y} = \{1, 2, \cdots, K\}$, where K is the number of classes. Conditional on the input x, Y follows a categorical distribution, and the conditional probability is denoted by

$$\boldsymbol{p}(\cdot|x) = [P(Y = 1|X = x), \cdots, P(Y = k|X = x)] \tag{2}$$

with the shorthand notation $p(x)_k = P(Y = k|X = x)$. If we knew $\boldsymbol{p}(\cdot|x)$, we could compute the conditional risk via

$$g(x) = \mathbb{E}[\ell(\hat{f}(X), Y)|X = x] = \sum_{k=1}^{K} \ell(\hat{f}(x), k)p(x)_k. \tag{3}$$

Regression. For regression problems, we have $\mathcal{Y} \subseteq \mathbb{R}$. We assume (X, Y) follows a general distribution on $P_{\mathcal{X}\times\mathcal{Y}}$, and conditional on $X = x$, we require that the conditional distribution $P_{\mathcal{Y}|x}$ admits a pdf $p(y|x)$. We have the following expression of $g(x)$ for regression problems:

$$g(x) = \mathbb{E}[\ell(\hat{f}(X), Y)|X = x] = \int_{\mathbb{R}} \ell(\hat{f}(x), y)p(y|x)dy. \tag{4}$$

2.1.1 Regression-based Approach for Conditional Risk Estimation

In this section, we illustrate that the conditional risk estimation problem can be formulated as a standard regression problem for both the classification and regression setting. We call this *regression-based* approach. With this formulation, the conditional risk calibration/estimation problem shares the same statistical guarantee as the standard regression problem.

Conditional Risk as Regressions. Denote by

$$\mathcal{G} = \{g_\theta | g_\theta : \mathcal{X} \to \mathbb{R}, \theta \in \Theta\}$$

the hypothesis class of all functions parametrized by θ in some parameter space Θ. For both the classification and regression problem, given the predictor $\hat{f}$,

the dataset $\{x_i, y_i\}_{i=1}^n$ sampled from $P_{\mathcal{X}\times\mathcal{Y}}$, and the loss function $L(\cdot,\cdot) : \mathcal{Y}\times\mathcal{Y} \to \mathbb{R}$, the machine learning problem aims to minimize

$$\min_{\theta\in\Theta} \mathbb{E}_{X\sim P_{\mathcal{X}}}[L(g_\theta(X), g(X))]. \tag{5}$$

A standard machine learning approach suggests treating $\{x_i, g(x_i)\}_{i=1}^n$ as the dataset and optimizing the empirical loss $\min_{\theta\in\Theta}\sum_{i=1}^n \frac{1}{n}L(g_\theta(x_i), g(x_i))$. However, computing $g(x_i)$ is often infeasible and we have to rely on its sample estimator $\ell(\hat{f}(x_i), y_i)$, for the reason that $\mathbb{E}[\ell(\hat{f}(x_i), y_i)] = g(x_i)$. As a result, to estimate the conditional loss, one practical approach is to train on the dataset $\{x_i, z_i\}_{i=1}^n$ with $z_i = \ell(\hat{f}(x_i), y_i)$, optimizing

$$\min_{\theta\in\Theta} \frac{1}{n}\sum_{i=1}^n L(g_\theta(x_i), z_i), \tag{6}$$

which is a regression problem. The standard generalization analyses shows that the generalization error of the regression problem depends on the complexity of the hypothesis class $\mathcal{G}$ for the predictor. Denote by

$$R(\theta) = \mathbb{E}[L(g_\theta(X), g(X))] \text{ and}$$
$$\hat{R}(\theta) = \frac{1}{n}\sum_{i=1}^n L(g_\theta(x_i), z_i)$$

the population risk and the sample risk, respectively, with θ^* and $\hat{\theta}$ the corresponding minimizers, and denote the empirical Rademacher complexity of $\mathcal{G}$ by

$$\hat{R}_n(\mathcal{G}) = \mathbb{E}\left[\sup_{\theta\in\Theta}\frac{1}{n}\sum_{i=1}^n \sigma_i g_\theta(x_i)\right],$$

where σ_i are i.i.d. with $P(\sigma_i = 1) = P(\sigma_i = -1) = 1/2$. We present the standard uniform convergence result on the generalization bound for L being the L^1 loss. The proof can be found in Appendix B.

Proposition 1 *Suppose $||\ell(\hat{f}(x), y)||_\infty < M$, with probability at least $1-\delta$, for the regression-based approach, we have the excess risk bounded by*

$$R(\hat{\theta}) - R(\theta^*) \le 2\hat{R}_n(\mathcal{G}) + 2M\sqrt{\frac{\ln(1/\delta)}{2n}}. \tag{7}$$

2.1.2 Calibration-based Approach for Conditional Risk Estimation

In this section, we present the *calibration-based* approach for the conditional risk calibration/estimation problem. Notice that this approach is *only* applicable for the classification setting. We show this approach features improved generalization error compared to (7), and more interpretable characterization of the conditional risk.

If we adopt another hypothesis class

$$\mathcal{H} = \{\boldsymbol{p}_\theta | \boldsymbol{p}_\theta : \mathcal{X} \to [0,1]^K, \theta\in\Theta\}$$

to parametrize the probability mass function $\boldsymbol{p}(\cdot|x)$ defined in (2), a natural estimator for $g(x)$ becomes

$$g_{ca,\theta}(x) = \sum_{k=1}^K \ell(\hat{f}(x), k)p_\theta(x)_k, \tag{8}$$

where the subscript means the calibration-based estimator for $g(\cdot)$. Next, the estimation of the conditional loss depends on how well we can approximate $\boldsymbol{p}(x)$ with $\boldsymbol{p}_\theta(x)$, which has its main interest in the domain of individual calibration (Zhao et al., 2020), that is we need $||\boldsymbol{p}_\theta(x) - \boldsymbol{p}(x)||$ to have some coverage guarantee for every $x \in \mathcal{X}$.

The following proposition establishes the connection between conditional risk estimation and individual calibration under the *weak realizability* condition.

Definition 1 *We say that the distribution $P_{\mathcal{X}\times\mathcal{Y}}$ and the function class $\mathcal{H}$ satisfy weak realizability for the classification problem if, for all $x \in \mathcal{X}$,*

$$[P(Y=1 \mid X=x), \cdots, P(Y=k \mid X=x)] \in \mathcal{H}.$$

An immediate result of the weak realizability assumption is that, for classification problems, the (unique) minimizer

$$\arg\min_{\boldsymbol{p}_\theta\in\mathcal{H}} \mathbb{E}_{X\sim P_{\mathcal{X}}}[\ell(\boldsymbol{p}_\theta(X), Y)] \tag{9}$$

can be used to construct minimizers for the conditional risk (5). The following proposition formalizes this result.

Proposition 2 *Under the weak realizability assumption, and for a suitable loss function ℓ such as the cross-entropy loss or Brier loss, we have*

$$\sum_{k=1}^K \ell(\hat{f}(\cdot), k)p_\theta(\cdot)_k \in \arg\min_{g_\theta\in\mathcal{G}} \mathbb{E}_{X\sim P_{\mathcal{X}}}[L(g_\theta(X), g(X))],$$

where $\boldsymbol{p}_\theta(\cdot)$ is the (unique) minimizer of (9).

Proposition 2 implies that in the classification setting, the conditional risk calibration problem can be stated as a calibration problem. Next, we provide related theoretical results for the excess risk, and denote

$$R_{ca}(\theta) = \mathbb{E}[L(g_{ca,\theta}(X), g(X))]$$

the population risk under the calibration-based approach. We refer the readers to Appendix B for the proof.

Theorem 1 *For K-class classification problems, suppose $||\ell(\hat{f}(x), y)||_\infty < M$, with probability at least $1-\delta$, we have the excess risk being*

$$R_{ca}(\hat{\theta}) - R_{ca}(\theta^*) \leq 2M\hat{R}_n(\mathcal{H}) + 2M\sqrt{\frac{\ln(1/\delta)}{2n}}. \quad (10)$$

Moreover, for population loss $R_{ca}(\theta)$, we have for all $\theta \in \Theta$

$$\begin{aligned} R_{ca}(\theta) &\leq M \cdot \mathbb{E}[||\boldsymbol{p}_\theta(X) - \boldsymbol{p}(X)||_1] \\ &\leq M\sqrt{K\left(2\hat{R}_n(\mathcal{H}) + \sqrt{\frac{\ln(1/\delta)}{2n}}\right)}. \end{aligned} \quad (11)$$

Theoretical Insights. In Theorem 1, the bound in (11) enables a direct control of the population risk $R_{ca}(\theta)$ for all $\theta \in \Theta$, implying a good calibration error results in better population risk bound. This yields stronger theoretical guarantees for the population risk $R_{ca}(\hat{\theta})$, where $\hat{\theta}$ is the empirical risk minimizer. Notably, under the typical classification setup, estimating the conditional probability distribution is often easier than estimating the conditional risk itself. This is because the probability distributions tend to exhibit more benign statistical properties, whereas the conditional risk depends intricately on the loss function $\ell(\cdot, \cdot)$, the predictor $\hat{f}$, and the underlying data distribution.

As a result, $R_{ca}(\theta^*)$ is usually lower than $R(\theta^*)$, which implies a lower population risk of $R_{ca}(\hat{\theta})$ compared to $R(\hat{\theta})$ (see (7) (10)). We formally put the above statement as a proposition with a concrete environment. This indicates calibration-based method's theoretical advantages over the regression-based method.

Proposition 3 *Assume the weak-realizability condition, and the distribution being perfectly separable; that is, there exists a function $h : \mathcal{X} \to \mathcal{Y}$ such that $\mathbb{P}(h(X) = Y) = 1$. Then for $\ell(\cdot, \cdot)$ being the 0-1 loss, and all predictor $\hat{f}$, $R_{ca}(\hat{\theta})$ has the upper bound of*

$$R_{ca}(\hat{\theta}) \leq 2\hat{R}_n(\mathcal{H}) + 2\sqrt{\frac{\ln(1/\delta)}{2n}}$$

In contrast, for regression-based methods, we have

$$R(\hat{\theta}) \leq R(\theta^*) + 2\hat{R}_n(\mathcal{G}) + 2\sqrt{\frac{\ln(1/\delta)}{2n}}$$

and there exists predictor $\hat{f}$ such that $R(\theta^*) > 0$.

In the experimental section, Proposition 3 is also empirically verified: the calibration-based method consistently achieves lower test empirical risk $\hat{R}(\hat{\theta})$, an unbiased estimator of the population risk $R(\hat{\theta})$, compared to regression-based methods.

3 EXPERIMENTS

In this section, we empirically verify the theoretical insights from the previous section. That is, consistent with Theorem 1, we show

- Calibration-based approach can provide better conditional risk estimation than the regression-based approach.
- Better calibration for probability function results in better conditional risk estimation.

Moreover, we also show that strong calibration risk estimation is beneficial for downstream applications. We conduct extensive experiments in the learning to defer application and observe strong empirical performance, achieving results that surpass previous benchmarks.

Regarding specific experimental goals, we outline the key questions investigated in our experimental section and provide general answers as follows:

- **What Improves Conditional Risk Estimation?** Several factors influence conditional risk estimation. First, recall that the conditional loss $g(x)$ in (1) is defined relative to the given predictor $\hat{f}$. A poorly performing predictor increases the scale of $z = \ell(\hat{f}(x), y)$, thereby deteriorating the estimation. Second, the choice of loss functions ℓ and L also affects the estimation process. Third, the performance of probability calibrator would also affect the quality of conditional risk estimation. Thus, it is important to disentangle all these factors, and understanding the performance of different estimation techniques across various predictors and loss functions is crucial.

  Our empirical results demonstrate that in the classification setting, calibration-based approaches consistently outperform regression-based methods across all settings. Furthermore, improved calibration techniques enhance conditional risk estimation for both weak and strong predictors, showing consistent improvements across different settings.

- **Implication for Downstream Tasks.** Another natural question to raise is that *does better conditional risk estimation imply better downstream task performance?* We conduct the empirical study for the learning to defer problem, both in the classification and regression setting. Our finding shows that the regression-based method is also effective for condition risk estimation, and strong conditional risk estimation can enhance the performance on learning to defer, both in the classifi-

cation setting and regression setting. For the particular task on regression with rejection, our proposed approach outperforms state-of-art benchmarks.

In Section 3.1, we study the classification setting, comparing the calibration-based method and regression-based method for conditional risk estimation. In Section 3.2, we study the classification and regression setting in the downstream application of learning to defer, and experiment both the calibration-based method and regression-based methods.

3.1 Conditional Risk Estimation for Classification

In this section, we investigate the performance of two methods – calibration-based and regression-based – for estimating the conditional risk $g(x)$ in the classification setting.

To show how strong and weak predictors and calibrators (estimators of the conditional risk) would affect the performance, we conduct experiments using three classifiers,

- $\hat{f}_1$, a CNN-based classifier
- $\hat{f}_2$, a ResNet-based classifier
- $\hat{f}_3$, a EfficientNet-based classifier

representing low, medium, and high performance tiers. The goal of our numerical study is to verify the theoretical findings of Propositions 1-2 and Theorem 1, which suggests that calibration-based methods outperform regression-based methods for all classifiers $\hat{f}_i$.

Calibration-based Methods. For a given classifier $\hat{f}$, its conditional loss $g(x)$ can be estimated by replacing the true probabilities $\boldsymbol{p}(x)$ in (3) with estimates $\boldsymbol{p}_\theta(x)$:

$$g_\theta(x) = \sum_{k=1}^{K} \ell(\hat{f}(x), k) p_\theta(x)_k \tag{12}$$

Following Proposition 2, we can fit probability estimators that minimize the cross-entropy loss or Brier loss. Specifically, to learn $\boldsymbol{p}_\theta(x)$, we train common computer vision models – CNN, ResNet, or EfficientNet – on the training data and their softmax probabilities would be $p_\theta(x)$. Thus, we can train calibrators $p_{\theta_1}(x)$, $p_{\theta_2}(x)$, and $p_{\theta_3}(x)$, corresponding to the probabilities of CNN, ResNet, and EfficientNet, respectively. Lastly, the calibration-based estimator is the plug-in estimator of (12).

Table 1: Aggregate Results for $\hat{f}_1$ (CNN) Across Different Calibration Approaches. The average loss of $\hat{f}_1$ is 1.30.

Loss	**Calibration-based**			**Regression-based**		
	$\hat{p}_{\theta_1}$	$\hat{p}_{\theta_3}$	$\hat{p}_{\theta_3,cali}$	$\hat{g}_{\theta_1}$	$\hat{g}_{\theta_1,rep}$	$\hat{g}_{\theta_3}$
L^1	0.96	**0.28**	1.71	2.36	1.09	2.77
L^2	4.06	**0.97**	5.13	10.86	4.07	15.10

Furthermore, after obtaining $p_{\theta_i}(x)$, we apply Platt scaling (Guo et al., 2017), a common calibration technique, to better reflect the true probability. We denote the resulting temperature-scaled probability estimates by $p_{\theta_i,cal}(x)$.

Regression-based Methods. Instead of using probability estimates, the regression-based method treats calibration of the conditional loss $g(x)$ as a standard regression problem. Since we focus on a computer vision task in this numerical experiment, we have to employ suitable architectures to learn $g(x)$. Again, we train three regression models $g_{\theta_1}(x)$, which is CNN-based, $g_{\theta_2}(x)$, which is ResNet-based, and $g_{\theta_3}(x)$, which is EfficientNet-based. Each model attaches a linear predictor to the last layer of the network and is trained from scratch.

In addition to training regressor from scratch, we also consider some "pre-trained" model, denoted by $g_{\theta,rep}(x)$. We first train some strong classifier with cross-entropy loss, then $g_{\theta,rep}(x)$ takes as input the final-layer representation from the classifier, and perform regression. Since training with cross-entropy loss maximizes the probability of the correct class, we expect that $g_{\theta,rep}(x)$ can benefit from more informed representations compared to $g_{\theta_i}(x)$ trained from scratch. We denote by $g_{\theta_i,rep}(x)$ the regressor utilizing the representations of $\hat{f}_i$ accordingly.

Results. We conduct experiment for this setting on CIFAR-10 dataset, where the number of classes K is 10. The results of our numerical study are presented in Tables 1-2. These tables compare the performance of different estimation approaches given various trained classifiers with respect to both L^1 and L^2 losses.

For calibration-based methods, we observe that better probability estimation – moving from the lower-performing CNN (p_{θ_1}) to ResNet (p_{θ_2}) and, ultimately, to EfficientNet (p_{θ_3}) – directly translates into lower calibration errors. This holds for all considered classifiers $\hat{f}$ and both loss functions. Interestingly, applying Platt scaling makes the calibration performance worse, probably due to the fact that Platt scaling is a group calibration technique instead of individual calibration, leading to distorted probability distributions in our ex-

Table 2: Aggregate Results for $\hat{f}_2$ (ResNet) Across Different Calibration Approaches. The average loss of $\hat{f}_2$ is 0.57.

Loss	Calibration-based			Regression-based		
	$\hat{p}_{\theta_2}$	$\hat{p}_{\theta_3}$	$\hat{p}_{\theta_3,cali}$	$\hat{g}_{\theta_2}$	$\hat{g}_{\theta_2,rep}$	$\hat{g}_{\theta_3}$
L^1	0.47	**0.36**	1.98	0.63	0.50	0.39
L^2	1.92	**1.40**	5.46	1.88	1.83	1.89

periment and results in worse performance.

The regression-based method in general have weak performance compared to the calibration-based method; Moreover, its performance for conditional risk estimation will be affected by the performance of the predictor $\hat{f}$. In our setting, $\hat{f}_1$ has sample loss 1.3 and $\hat{f}_2$ has sample loss 0.57, and the conditional risk estimation is better for $\hat{f}_2$. These results are, again, consistent with our theoretical findings. More advanced architectures also lead to reduced loss for estimating conditional risk. Furthermore, using representations from pre-trained classifiers improves performance over training from scratch.

Overall, EfficientNet probability calibration (p_{θ_3}) without Platt scaling leads to the best performance across all considered calibration approaches.

3.2 Learning to Defer

In this section, we start with the regression setting of the learning to defer (L2D) problem. The regression setting of the L2D problem is also known as regression with rejection. We start with the regression setting, since Section 3.1 already provides enough supportive results for the classification setting. After covering the regression with rejection setting, we turn to the classification setting of the L2D problem and present similar results. Due to the space constraint, for the classification setting, we put the results in the appendix.

3.2.1 Regression with Rejection

Regression with rejection (RwR) extends the traditional regression framework by introducing a rejector mechanism. Given an input x, the system first evaluates $g(x)$, which quantifies the confidence in the regressor's prediction $f(x)$. If the confidence is sufficiently high, the system outputs $f(x)$; otherwise, it defers x to a human expert, who provides a more reliable prediction but at an additional cost c.

This approach is particularly valuable in scenarios where incorrect predictions can result in severe losses, and human expertise may be leveraged for uncertain cases. While the majority of research on learning with rejection has focused on classification problems, there is a growing body of literature exploring regression with rejection (Wiener and El-Yaniv, 2012; Geifman and El-Yaniv, 2019b; Jiang et al., 2020a; Zaoui et al., 2020a; Shah et al., 2022a; Cheng et al., 2024a; Li et al., 2024). We now formally define the RwR problem under the fixed-cost setting.

Consider a dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ drawn independently from an unknown distribution $\mathcal{P}_{\mathcal{X}\times\mathcal{Y}}$. The RwR problem involves learning two components: a regressor $f : \mathcal{X} \to \mathcal{Y}$ and a rejector $r : \mathcal{X} \to \{0, 1\}$. If $r(x) = 1$, the regressor's prediction for sample x is accepted, and if $r(x) = 0$, the sample is deferred to a human expert incurring a fixed cost of $c > 0$. The RwR loss function is designed to balance prediction errors and rejection costs. For example, let $\ell(\cdot, \cdot)$ be the standard L^2 loss, $\ell(\hat{y}, y) = (\hat{y} - y)^2$. Then the RwR loss function is formulated as:

$$\ell_{\text{RwR}}(f, r; x, y) = r(x) \cdot \ell(f(x), y) + (1 - r(x)) \cdot c \quad (13)$$

Let $\mathcal{G}$ and $\mathcal{R}$ be the sets of candidate regressors and rejectors. The objective of the RwR problem is to find the optimal regressor-rejector pair (f, r) that minimizes the expected RwR loss (13):

$$\min_{f\in\mathcal{G},\, r\in\mathcal{R}} L_{\text{RwR}}(f, r) = \mathbb{E}_{(X,Y)\sim\mathcal{P}_{\mathcal{X}\times\mathcal{Y}}} \left[\ell_{\text{RwR}}(f, r; X, Y)\right]. \quad (14)$$

To show the effectiveness of conditional risk calibration, we focus on a particular RwR setting called *no-rejection learning* (Li et al., 2024), where the regressor is trained on all available samples as in a standard regression task, and the rejector is learned separately using conditional risk calibration. *No-rejection learning* is a well-studied framework with theoretical guarantee that the regressor can achieve optimal performance on the RwR task when complemented by a suitable rejector. Specifically, suppose that the rejector $r(x)$ for a trained regressor $\hat{f}$ is defined with the help of a trained conditional risk calibrator $g_\theta(x)$ as follows:

$$r_\theta(x) := \mathbb{1}\{g_\theta(x) \le c\}, \quad (15)$$

and the calibrator $g_\theta(x)$ in (15) is trained to predict the L^2 conditional loss of $\hat{f}$, then the excess RwR loss can be upper-bounded with the following guarantees.

Proposition 4 (Li et al., 2024) *For a given $\hat{f}$ and the calibrator $g_\theta(x)$ that predicts the L^2 conditional loss of $\hat{f}$:*

$$L_{RwR}(\hat{f}, r_\theta) - L_{RwR}(f^*, r^*) \le \mathbb{E}\Big[(\hat{f}(X) - f^*(X))^2\Big] + \mathbb{E}\Big[|g_\theta(X) - g(X)|\Big],$$

where (f^*, r^*) denotes the optimal regressor-rejector pair that minimizes (14).

These results motivate the application of conditional risk calibration techiniques in Section 2.1 for the RwR problem (14) via a no-rejection learning strategy in (15). In this work, we perform systematic experiments to evaluate different conditional risk calibration strategies under different regressors. We demonstrate the benefits of conditional risk calibration for the RwR task by identifying methods that outperform prominent benchmarks across multiple open-source datasets. The remainder of this section conveys the following key messages:

- Improved conditional risk calibration leads to better RwR performance for both weak and strong regressors.
- Leveraging this insight, we compare our approach with previous RwR algorithms and consistently achieve better performance.

Experimental Setup. We conduct experiments on 8 UCI datasets (Kelly et al., 2023), which is a common experiment setup (Li et al., 2024). For each dataset, we considered 4 fixed rejection costs $c \in \{0.2, 0.5, 1.0, 2.0\}$, resulting in multiple RwR settings defined by the dataset-cost combinations. We then evaluated 16 methods for each RwR setting by training all possible regressor-calibrator pairs $(\hat{f}, g_\theta)$. We train $\hat{f}$ and g_θ using different models, ranging from simple models with relatively poor performance to more complex models that achieve better accuracy:

- LR: Linear regression;
- RF: Random forest;
- MLP: Multilayer perceptron;
- MLP2: Multilayer perceptron with two hidden layers.

Since the UCI datasets for RwR tasks do not contain images, we choose not to use ResNet or CNN models, as we did in active learning. It is generally accepted that for regression tasks, RF-based models and NN-based models perform relatively well.

All models, except for the random forest, were trained using standardized feature data. We performed 10-fold cross-validation in our experiments. In each fold, the dataset was partitioned into three subsets: 50% of the data were used to train the regressor $\hat{f}$, 40% to train the calibrator g_θ, and the remaining 10% were reserved for testing. The full results of our numerical experiments, along with further implementation details, are provided in Appendix E.

Our systematic study provides insights into the benefits of improved conditional risk calibration and identifies methods that outperform existing benchmarks in the same RwR setting.

Table 3: Absolute Errors of Calibrators Given Pretrained Predictors.

Data	Predictor	Predictor Squared Loss	Calibrator Absolute Error			
			LR	MLP	MLP2	RF
Concrete	**LR**	112.4	107.19	96.76	99.99	**75.37**
	MLP	83.57	80.6	73.65	77.07	**61.01**
	MLP2	58.77	59.76	55.07	56.61	**47.94**
	RF	33.15	37.99	35.98	36.38	**32.26**
Wine	**LR**	0.36	0.37	0.4	0.41	**0.34**
	MLP	0.25	0.26	0.26	0.28	**0.24**
	MLP2	0.26	0.32	0.34	0.35	**0.3**
	RF	0.3	0.36	0.38	0.4	**0.32**
Airfoil	**LR**	23.55	23.49	22.72	19.2	**13.53**
	MLP	14.01	14.32	13.41	12.25	**9.2**
	MLP2	6.28	6.75	6.37	5.95	**5.23**
	RF	4.62	5.29	5.08	4.81	**4.24**
Energy	**LR**	8.24	7.87	7.65	4.28	**2.41**
	MLP	7.54	6.74	6.53	3.67	**2.21**
	MLP2	2.36	2.01	1.91	1.64	**1.17**
	RF	0.32	0.36	0.35	0.4	**0.31**
Housing	**LR**	24.96	31.73	26.28	25.53	**24.23**
	MLP	18.06	26.89	**20.57**	22.92	20.91
	MLP2	12.56	17.33	14.53	17.29	**14.18**
	RF	14.56	20.44	17.47	19.04	**17.3**
Solar	**LR**	0.63	**0.81**	0.91	0.96	0.85
	MLP	0.66	**0.86**	0.94	1	0.88
	MLP2	0.7	**0.92**	0.99	1.04	0.94
	RF	0.81	**1.05**	1.12	1.21	1.07
Forest	**LR**	2.11	**2.02**	2.19	2.76	2.22
	MLP	2.09	**2.02**	2.12	2.86	2.18
	MLP2	2.12	**2.03**	2.15	2.96	2.19
	RF	2.22	**2.23**	2.33	3.06	2.35
Parkinsons	**LR**	86.53	76.24	66.75	37.07	**21.23**
	MLP	19.87	21.4	19.43	18.68	**16.17**
	MLP2	6.62	8.82	7.82	8.54	**7.07**
	RF	0.35	0.57	0.75	0.74	**0.45**

Model Performance for Predictors and Calibrators. In terms of interpretation of the results, we adapt a naming scheme where each model was denoted by the combination of its regressor and calibrator. For example, MLP+RF indicates that the regressor $\hat{f}$ is a multi-layer perceptron with one hidden layer, and the calibrator g_θ is a random forest.

The first step is to identify strong and weak predictors and calibrators among our models. Table 3 indicates that LR generally performs relatively poorly as both a predictor and a calibrator, whereas RF and MLP2 exhibit comparatively better performance in both roles. The only exceptions are the Solar and Forest datasets, where LR demonstrates strong performance due to their target distributions, where a single target value dominates the data – especially in the case of Solar, where 83% of targets correspond to just one value. In cases where all models yield similar estimations (e.g., Wine, Solar, and Forest), the predictor loss and calibrator loss are relatively low in scale compared to other

datasets. This analysis enables a general categorization of the four models based on their effectiveness as predictors and calibrators.

Table 4: Relationship Between Calibrator Performance and RwR Loss for the ENERGY Dataset.

	Calibrator Absolute Error	RwR Loss			
		$c = 0.2$	$c = 0.5$	$c = 1.0$	$c = 2.0$
LR+LR	**7.87**	0.77	1.03	1.46	2.17
LR+MLP	**7.65**	0.52	0.79	1.20	1.93
LR+MLP2	**4.28**	0.31	0.61	1.03	1.62
LR+RF	**2.41**	0.26	0.54	0.85	1.29
MLP+LR	**6.74**	0.44	0.69	1.06	1.65
MLP+MLP	**6.53**	0.43	0.69	1.18	1.97
MLP+MLP2	**3.67**	0.27	0.56	0.98	1.60
MLP+RF	**2.21**	0.20	0.46	0.80	1.32
MLP2+LR	**2.01**	0.23	0.49	0.79	1.18
MLP2+MLP	**1.91**	0.23	0.49	0.77	1.15
MLP2+MLP2	**1.64**	0.27	0.50	0.78	1.13
MLP2+RF	**1.17**	0.20	0.45	0.71	1.04
RF+LR	**0.36**	0.17	0.28	0.31	0.33
RF+MLP	**0.35**	0.16	0.27	0.32	0.32
RF+MLP2	**0.40**	0.17	0.28	0.32	0.34
RF+RF	**0.31**	0.14	0.26	0.30	0.32

Relationship Between Calibration Quality and RwR Performance. Our study also showed that, within a group of calibration methods under a fixed RwR setting, better calibration leads to improved RwR performance. Table 4 presents the results for the ENERGY dataset across different costs. In particular, for a fixed RwR setting and regressor, we computed the mean absolute errors and the RwR losses that correspond to each calibrator, and examined the relationship between these two metrics. To facilitate comparison, we highlighted lower values in blue and higher values in red within each group of calibrators (sharing the same RwR setting and regressor), making it easy to observe the relationship between calibrators' errors and the RwR losses. This example clearly demonstrates that lower calibrator errors generally correspond to lower RwR costs. This finding further justifies our study on calibrating conditional risk for various downstream tasks, also corroborating with the theoretical findings of Li et al. (2024).

More specifically, the RF calibrator outperformed other calibrators for the vast majority of the considered datasets (see Table 3). To illustrate this performance, we present Figure 1 in Appendix E for the results of using the RF calibrator with a trained MLP regressor on the ENERGY test data (from one fold). This plot shows the true target values $f(x)$, the predictions made by the MLP regressor $\hat{f}(x)$, and the square root of the losses estimated by the RF calibrator $g_\theta(x)$. The test instances are sorted in ascending order of the target value to improve clarity. By comparing the calibrator's estimates with the sample conditional risk, we observe that the calibrator performs exceptionally well, accurately reflecting the MLP prediction errors.

Advantage Over Existing Results. According to Table 4, the two methods that showed the best overall performance in our analysis are MLP+RF and RF+RF. In Table 6 (Appendix E.3), we compare the performance of these methods against the following two RwR benchmarks with respect to the RwR loss (13): SelectiveNet (Geifman and El-Yaniv, 2019b) and NN+kNNRej (Li et al., 2024). SelectiveNet is a deep learning approach with an integrated rejector, whereas NN+kNNRej is a *no-rejection learning* algorithm that demonstrated the best performance in the study by Li et al. (2024).

Table 6 (in Appendix E.3) shows that MLP+RF matches or surpasses NN+kNNRej in 30 out of 32 RwR settings, and in 27 out of 32 settings when compared to SelNet. In the remaining cases, the differences in RwR losses are generally small. Similarly, RF+RF matches or surpasses NN+kNNRej in 24 out of 32 settings, and in 25 out of 32 settings when compared to SelNet. It is worth noting that for the PARKINSONS dataset, the RF+RF method substantially outperformed the other methods, which can be explained by the superior performance of the regressor.

Conclusion. From the progressive experiment design in Section 3.2.1, we observe that better conditional risk calibration can improve the RwR loss for both strong and weak predictors; by adopting a better calibrator for the conditional risk, our method outperforms the current benchmark in the RwR task.

3.2.2 Classification Setting for L2D

Due to the space limitation, we put the discussion of the classification setting in Appendix C.

4 CONCLUSION

In this work, we formulate and analyze the problem of conditional risk calibration, establishing its theoretical foundations and practical significance. Our theoretical analyses are supported by empirical studies, which validated our insights and demonstrated the effectiveness of improved conditional risk estimation and its downstream application. By bridging theoretical understanding with real-world applications, our work lays the foundation for further advancements in uncertainty-aware decision-making and the development of more reliable, interpretable, and risk-aware machine learning systems.

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Checklist

From the authors: our responses are marked in green.

1. For all models and algorithms presented, check if you include:
   (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes/No/Not Applicable]
   (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes/No/Not Applicable]
   (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Yes/No/Not Applicable]

2. For any theoretical claim, check if you include:
   (a) Statements of the full set of assumptions of all theoretical results. [Yes/No/Not Applicable]
   (b) Complete proofs of all theoretical results. [Yes/No/Not Applicable]
   (c) Clear explanations of any assumptions. [Yes/No/Not Applicable]

3. For all figures and tables that present empirical results, check if you include:
   (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [Yes/No/Not Applicable]
   (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes/No/Not Applicable]
   (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes/No/Not Applicable]
   (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes/No/Not Applicable]

4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
   (a) Citations of the creator If your work uses existing assets. [Yes/No/Not Applicable]
   (b) The license information of the assets, if applicable. [Yes/No/Not Applicable]
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   (d) Information about consent from data providers/curators. [Yes/No/Not Applicable]
   (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Yes/No/Not Applicable]

5. If you used crowdsourcing or conducted research with human subjects, check if you include:
   (a) The full text of instructions given to participants and screenshots. [Yes/No/Not Applicable]
   (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Yes/No/Not Applicable]
   (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Yes/No/Not Applicable]

Calibrating Conditional Risk: Supplementary Materials

A RELATED WORK

Uncertainty Estimation While this conditional risk estimation/calibration problem has not been explicitly studied in prior work, it is closely related to established areas such as uncertainty quantification and calibration. One of the most common approaches to uncertainty quantification is probability calibration, where a model's predicted probabilities are adjusted to reflect true likelihoods. Methods such as Platt scaling (Platt et al., 1999) , isotonic regression (Zadrozny and Elkan, 2002), and temperature scaling (Guo et al., 2017) have been widely used to improve the reliability of predicted probabilities. However, these methods focus on calibrating probabilities rather than directly estimating the expected loss, making them insufficient for conditional risk estimation. Recent work on conformal prediction (Angelopoulos et al., 2023) and individual calibration Zhao et al. (2020) has explored methods for instance-level uncertainty estimation, but these approaches typically rely on confidence scores rather than explicit loss estimation.

Human-AI Interaction Our study is related to the Learning to Defer (L2D) setting (Madras et al., 2018; Verma and Nalisnick, 2022; Mozannar et al., 2023), which belong to the general field of Human-AI Interaction. Similar problems have been studied under related frameworks, including learning under triage (Okati et al., 2021), learning under human assistance (De et al., 2020), learning to complement humans (Wilder et al., 2020), and selective prediction (Geifman and El-Yaniv, 2019a), all of which share a similar setup.

Learning to Defer We conduct experiment setting when the cost of deferral is constant, which has a long history in machine learning and goes by the name of rejection learning (Chow, 1970; Bartlett and Wegkamp, 2008; Cortes et al., 2016). The more recent development involves various aspects including the one-stage vs two-stage process (Raghu et al., 2019; Mao et al., 2024a), mixture of experts (Pradier et al., 2021; Wilder et al., 2020), and the related surrogate loss (Charusaie et al., 2022; Verma and Nalisnick, 2022; Keswani et al., 2021; Mao et al., 2024b).

Regression with Rejection The Regression with Rejection is a special setting of L2D. In this setting Zaoui et al. (2020b) characterize the optimal predictor and rejector for the function class of all measurable functions, and propose a nonparametric algorithm to learn them. Kang and Kang (2023) quantify the prediction uncertainty directly, which can also be applied to address the regression with rejection problem. Jiang et al. (2020b) aim to minimize the rejection rate under a specified level of loss without considering rejection costs. Shah et al. (2022b) consider both rejection costs and reject budgets, highlighting the challenges of optimizing the predictor and rejector even for training samples, and develop a greedy algorithm to solve the problem approximately. Also see Li et al. (2024); Cheng et al. (2024b) for more recent developments.

B PROOFS

B.1 Proof of Proposition 1

From the standard telescoping sum, we have

$$\begin{aligned} R(\hat{\theta}) - R(\theta^*) &= R(\hat{\theta}) - \hat{R}(\hat{\theta}) + \hat{R}(\hat{\theta}) - \hat{R}(\theta^*) + \hat{R}(\theta^*) - R(\theta^*) \\ &\leq R(\hat{\theta}) - \hat{R}(\hat{\theta}) + \hat{R}(\theta^*) - R(\theta^*) \\ &\leq 2 \sup_{\theta \in \Theta} |R(\theta) - \hat{R}(\theta)|. \end{aligned}$$

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This leads to the uniform convergence of the generalization error, which is a standard approach in statistical learning textbook. By the standard symmetrization trick (Mohri, 2018), we have that

$$\mathbb{E}\left[\sup_{\theta\in\Theta}|R(\theta)-\hat{R}(\theta)|\right] \leq 2\mathbb{E}[\hat{R}_n(\mathcal{L})] \leq 2\mathbb{E}[\hat{R}_n(\mathcal{G})].$$

For the above inequality, $\mathcal{L}$ is the loss function class $\mathcal{L} = \{L(\cdot,\cdot) : \mathcal{X}\times\mathcal{Y} \to \mathbb{R} \,|\, L(x,z) = |g_\theta(x) - z|$, where $g_\theta \in \mathcal{G}\}$. Since the function class $\mathcal{L}$ is direcly related with $\mathcal{G}$, we have that the function L is 1-Lipchitz with respect to $g_\theta(x)$, therefore, we have $\mathbb{E}[\hat{R}_n(\mathcal{L})] \leq 1\cdot\mathbb{E}[\hat{R}_n(\mathcal{G})]$, where $\mathbb{E}[\hat{R}_n(\mathcal{G})]$ is the the empirical Rademacher complexity defined as $\hat{R}_n(\mathcal{G}) = \mathbb{E}_\sigma\left[\sup_{\theta\in\Theta}\frac{1}{n}\sum_{i=1}^n \sigma_i g_\theta(x_i)\right]$ where σ_i are i.i.d. with $P(\sigma_i = 1) = P(\sigma_i = -1) = 1/2$. Knowing that $||\ell(\hat{f}(x), y)||_\infty < M$, a standard bounded-difference or Hoeffding-type inequality with a union bound ensures with probability at least $1-\delta$, we have

$$\sup_{\theta\in\Theta}|R(\theta)-\hat{R}(\theta)| \leq 2\hat{R}_n(\mathcal{G}) + M\sqrt{\frac{\ln(1/\delta)}{2n}}.$$

B.2 Proof of Proposition 2

By weak realizability, the probability $\boldsymbol{p}(x)$ is in the class $\mathcal{H}$. Because we assume ℓ to be cross-entropy or Brier score, which are proper classification loss. It is well-known that under the proper loss, $\boldsymbol{p}(x)$ uniquely minimizes the pointwise expected loss such that

$$\boldsymbol{p}(x) \in \arg\min_{\boldsymbol{p}_\theta(x)\in\mathcal{H}} \mathbb{E}_{X\sim P_\mathcal{X}}[\ell(\boldsymbol{p}_\theta(X), Y)].$$

Consequently, we denote $\boldsymbol{p}_{\theta*}(x) = \boldsymbol{p}(x)$ as the minimizer. $\boldsymbol{p}_\theta$ induces a function

$$g_{\theta^*}(x) := \sum_{k=1}^K \ell(\hat{f}(x), k)p_{\theta^*}(x)_k.$$

Also notice that $g_{\theta^*}(x)$ exactly coincides with the true conditional loss $g(x)$ because

$$g_\theta(x) = \sum_{k=1}^K \ell(\hat{f}(x), k)p_\theta(x)_k.$$

Therefore, we have $L(g_{\theta^*}(x), g(x)) = 0$ for all $x \in \mathcal{X}$. Lastly, it follows that $\mathbb{E}_{X\sim P_\mathcal{X}}[L(g_{\theta^*}(X), g(X))] = 0$, reaching the minimum for any standard loss function L. Therefore, we have

$$\sum_{k=1}^K \ell(\hat{f}(\cdot), k)p_{\theta^*}(\cdot)_k = g_{\theta^*} \in \arg\min_{g_\theta\in\mathcal{G}} \mathbb{E}_{X\sim P_\mathcal{X}}[L(g_\theta(X), g(X))].$$

B.3 Proof of Theorem 1

The proof of (10) is the same as that of Proposition 1. One just have to realize that in the classfication setting, we have

$$g_{ca,\theta}(x) \leq \sum_{k=1}^K |\ell(\hat{f}(x), k)|p_\theta(x)_k \leq M\sum_{k=1}^K p_\theta(x)_k = M.$$

For (11), notice that the first inequality can be proved by

$$|g_{ca,\theta}(x) - g(x)| = \left|\sum_{k=1}^K \ell(\hat{f}(x), k)(p_\theta(x)_k - p(x)_k)\right| \leq M\sum_{k=1}^K |p_\theta(x)_k - p(x)_k| = M||\boldsymbol{p}_\theta(x) - \boldsymbol{p}(x)||_1.$$

For the second inequality, it suffices to bound $||\boldsymbol{p}_\theta(x) - \boldsymbol{p}(x)||_1$ by some form of loss $R(\theta) - R(\theta^*)$, and then apply the uniform bound.

First, to incorporate the risk, we need to work on the L^2 distance instead of L^1. We apply the bound $||\boldsymbol{p}_\theta(x) - \boldsymbol{p}(x)||_1 \leq \sqrt{K}||\boldsymbol{p}_\theta(x) - \boldsymbol{p}(x)||_2$ and Jensen's inequality $\mathbb{E}[||\boldsymbol{v}||_2] \leq \sqrt{\mathbb{E}[||\boldsymbol{v}||_2^2]}$ such that

$$\mathbb{E}[||\boldsymbol{p}_\theta(X) - \boldsymbol{p}(X)||_1] \leq \sqrt{K\mathbb{E}[||\boldsymbol{p}_\theta(X) - \boldsymbol{p}(X)||_2^2]}.$$

Next, we incorporate the population risk. Consider the specifically defined loss function $\ell(\boldsymbol{p}_\theta(x), \boldsymbol{y}) = ||\boldsymbol{p}_\theta(x) - \boldsymbol{y}||_2^2$, where $\boldsymbol{y}$ is the one-hot vector for the corresponding label of x. Then we define

$$R(\theta) = \mathbb{E}[||\boldsymbol{p}_\theta(x) - \boldsymbol{y}||_2^2],$$

Note that since the weak-realizability condition is assumed throughout, we take the convention that there exists $\theta^* \in \Theta$ such that $\boldsymbol{p}_{\theta^*}(x) = \boldsymbol{p}(x)$, and denote $R(\theta^*) = \mathbb{E}[||\boldsymbol{p}(x) - \boldsymbol{y}||_2^2]$.

Then, from geometry property of the L^2 distance

$$||\boldsymbol{p}_\theta(x) - \boldsymbol{y}||_2^2 = ||\boldsymbol{p}_\theta(x) - \boldsymbol{p}(x)||_2^2 + ||\boldsymbol{p}(x) - \boldsymbol{y}||_2^2 + (\boldsymbol{p}_\theta(x) - \boldsymbol{p}(x))^\top(\boldsymbol{p}(x) - \boldsymbol{y})$$

and the property that $\mathbb{E}[\boldsymbol{p}(x) - \boldsymbol{y}] = 0$, we know

$$||\boldsymbol{p}_\theta(x) - \boldsymbol{p}(x)||_2^2 = ||\boldsymbol{p}_\theta(x) - \boldsymbol{y}||_2^2 - ||\boldsymbol{p}(x) - \boldsymbol{y}||_2^2,$$

establishing the identity that

$$\mathbb{E}[||\boldsymbol{p}_\theta(x) - \boldsymbol{p}(x)||_2^2] = R(\theta) - R(\theta^*). \tag{16}$$

Now we are left with

$$\mathbb{E}[R_{ca}(\theta)] \leq M\,\mathbb{E}[||\boldsymbol{p}_\theta(x) - \boldsymbol{p}(x)||_1] \leq M\sqrt{K(R(\theta) - R(\theta^*))},$$

and the excess risk $R(\theta) - R(\theta^*)$ can be bounded by the same uniform convergence argument in Proposition 1, and the estimation target is a probability function bounded by 1. By repeating the same trick, we can have a bound of $2\hat{R}_n(\mathcal{H}) + \sqrt{\frac{\ln(1/\delta)}{2n}}$.

B.4 Proof of Proposition 3

For the calibration-based method, from the weak-realizability condition and perfectly separable condition, we know that for every sampled x, there must be one corresponding class (implied by the separability condition), denoted by y_x, and we have the property that $\boldsymbol{p}_{\theta^*}(x) = \boldsymbol{p}(x)$ (implied by the weak-realizability condition). Next, we have

$$g_{ca,\theta^*}(x) = \sum_{k=1}^{K} \ell(\hat{f}(x), k)p_{\theta^*}(x)_k = \ell(\hat{f}(x), y_x), \quad \text{and}$$
$$g(x) = \sum_{k=1}^{K} \ell(\hat{f}(x), k)p(x)_k = \ell(\hat{f}(x), y_x).$$

Notice that this is independent of $\hat{f}$, hence having $g(x) = g_{ca,\theta^*}(x)$ for all $\hat{f}$. Therefore, for the population risk,

$$\begin{aligned} R_{ca}(\theta^*) &= \mathbb{E}[(g_{ca,\theta^*}(X) - g(X))^2] \\ &= \int (\ell(\hat{f}(x), y_x) - \ell(\hat{f}(x), y_x))^2 p(x)dx = 0. \end{aligned}$$

Then the first bound is proved by (10).

For the regression-based method, it suffices to prove the existence of $\hat{f}$ such that for its corresponding conditional loss $g(x) = \mathbb{E}[\ell(\hat{f}(x), Y)|X = x]$, we have

$$R(\theta^*) = \inf_{\theta\in\Theta} R(\theta) = \inf_{\theta\in\Theta} \mathbb{E}[|g(X) - g_\theta(X)|] > 0.$$

Without loss of generality, we can prove this by considering a binary classification problem with $\hat{f}(x)$ being the indicator function of the *fat cantor set* $\mathbb{C}$ (Aliprantis and Burkinshaw, 1998) such that

$$\hat{f}(x) = \begin{cases} 1 & \text{for } x \in \mathbb{C} \\ 0 & \text{for } x \notin \mathbb{C}. \end{cases}$$

Note that $\mathbb{C}$ is uncountable, nowhere dense, but have positive measure. Therefore, because the data is continuously distributed, there must exists a region D with positive support such that without loss of generality for all $x \in D$, $y_x = 1$. Henceforth, we have

$$g(x) = \mathbb{E}[\ell(\hat{f}(x), Y)|X = x] = \sum_{k=0}^{1} \ell(\hat{f}(x), k)p_k(x) = \ell(\hat{f}(x), y_x),$$

which implies for $x \in D$

$$g(x) = \begin{cases} 0 & \text{for } x \in \mathbb{C} \\ 1 & \text{for } x \notin \mathbb{C}. \end{cases}$$

Notice that now the function $g(x)$ does not belong to any continuous hypothesis class $\mathcal{G}$ and therefore there exists a $\bar{\delta} > 0$ such that $\inf_{\theta \in \Theta} \int_D |g(x) - g_\theta(x)|dx \geq \bar{\delta}$. Since D has positive support, there exists $\delta > 0$ such that

$$R(\theta^*) = \inf_{\theta \in \Theta} \mathbb{E}[|g(X) - g_\theta(X)|] \geq \inf_{\theta \in \Theta} \mathbb{E}[|g(X) - g_\theta(X)| \cdot 1_D] > \delta.$$

Combing with (7) finish the second part of the proof.

C Classification Setting for L2D

Continuing the discussion in Section 3.2, we conduct the L2D experiment for the classification setting. The loss will be on the standard L2D loss (13) and (15), with the predictor being a classification model instead.

Experimental Setup. We employ two predictors of increasing performance: a CNN model as the weak predictor, a ResNet model as the strong predictor. In terms of the calibrator, similar to Section 3.1, we adopt both calibration-based method and regression-based method. For every predictor, we train three calibration-based methods, with the estimated probability taken from the logit probability of the CNN model, ResNet model, and EfficientNet model, respectively. In terms of the regression-based model, based on the experiment in Table 1 and 2, we extract representations from an EfficientNet classifier as the input for the regressor to estimate the conditional risk.

We present the calibrator's absolute error (relative to the true loss) and the corresponding L2D loss in Table 5 for various predictor–calibrator combinations. Similar to the findings in previous sections, for both the CNN and ResNet predictors, using better calibration models leads to a reduction in L2D loss. Lastly, the representation-based regression approach performs poorly in terms of both calibrator absolute error and L2D loss.

Table 5: Relationship Between Calibrator Performance and L2D Loss for the CIFAR-10 Dataset. Here, EFFNET stands for EfficientNet.

	Calibrator Absolute Error	**L2D Loss**			
		$c = 0.2$	$c = 0.5$	$c = 1.0$	$c = 2.0$
CNN+CNN	**0.76**	0.20	0.41	0.65	0.75
CNN+RESNET	**0.45**	**0.15**	**0.32**	**0.51**	**0.70**
CNN+EFFNET	**0.47**	0.16	**0.32**	0.53	0.74
CNN+FEATURE	**1.49**	0.28	0.47	0.67	0.82
RESNET+CNN	**1.48**	0.17	0.36	0.60	0.88
RESNET+RESNET	**0.48**	0.22	0.38	0.53	**0.57**
RESNET+EFFNET	**0.62**	**0.13**	**0.26**	**0.42**	0.61
RESNET+FEATURE	**1.03**	0.25	0.40	0.56	0.68

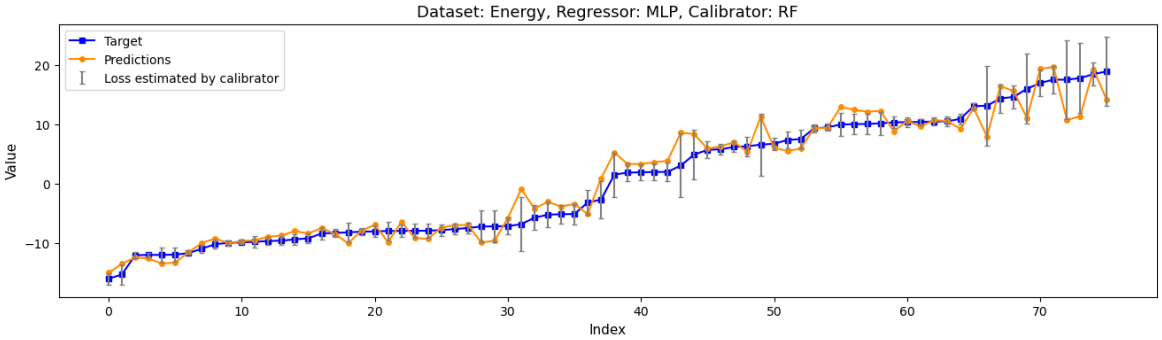

Figure 1: MLP Losses Estimated by an RF Calibrator on the ENERGY Test Data.

D CLASSIFICATION EXPERIMENT DETAILS

For $\hat{f}_1$, a CNN-based classifier, we employ a minimal convolutional neural network for CIFAR-10 classification. The network consists of two convolutional blocks followed by two fully connected layers. In the first block, a 3×3 convolution with 16 filters (stride 1, padding 1) is applied to the 32×32×3 input, followed by a ReLU activation and a 2×2 max pooling operation that halves the spatial resolution to 16×16. In the second block, a 3×3 convolution with 32 filters (stride 1, padding 1) is used, again followed by ReLU and a 2×2 max pooling, reducing the spatial dimensions to 8×8. The resulting feature map (of size 32×8×8) is flattened and fed into a fully connected layer with 64 units (with ReLU activation), followed by a final fully connected layer that produces logits for 10 classes.

For $\hat{f}_2$, a ResNet-based classifier, we utilize the ResNet-8 model, which is a compact residual network designed for CIFAR-10 classification. It begins with a 3×3 convolution that maps the 3-channel input to 16 feature maps, followed by batch normalization and a ReLU activation. This is followed by three residual layers, each consisting of a single basic block. Each basic block contains two 3×3 convolutional layers (with batch normalization and ReLU activations) and a shortcut connection that either passes the input unchanged or applies a 1×1 convolution (with batch normalization) to match dimensions when necessary. In our configuration, the first block preserves the spatial resolution, while the second and third blocks use a stride of 2 in their first convolution to downsample the feature maps. Finally, a global average pooling layer aggregates the spatial information, and a fully connected layer produces the final class scores. The network comprises eight layers: one initial convolution, six convolutional layers within the residual blocks, and one final fully connected layer.

For $\hat{f}_3$, a EfficientNet-based classifier, we initialize an EfficientNet-B0 model pre-trained on ImageNet (using the EfficientNet_B0_Weights.IMAGENET1K_V1). The default classifier in EfficientNet-B0 is a sequential block containing a dropout layer followed by a linear layer that maps a 1280-dimensional feature vector to 1000 classes. To adapt the model for CIFAR-10, we replace the final linear layer with a new one that outputs 10 classes.

In order to use representations as input for regressing conditional risk using a computer vision regresser (say CNN), we extract representations from an computer vision classifier (say EfficientNet) by removing its final linear layer, yielding a 1280-dimensional embedding, and train a simple multilayer perceptron (with one hidden layer of 256 units) to regress the cross-entropy loss of the CNN prediction model. The loss function in training is the mean squared error loss. We then use this calibrated loss to guide our decisions in L2D setting.

We partition the CIFAR-10 dataset into three segments: 50% of the data is used to train the classification predictors, 40% is dedicated to training the feature representation and the MLP loss predictor, and the remaining 10% is reserved for testing the L2D loss. Standard data augmentations—including random cropping, horizontal flipping, and mean normalization—are applied during training, which is conducted using an early stopping strategy with a five-round patience. The predictors are optimized using cross-entropy loss and the AdamW optimizer, with a learning rate of 1×10^{-3} and a weight decay of 1×10^{-4}. The training accuracy for the three predictors $\hat{f}_1$, $\hat{f}_2$, $\hat{f}_3$ are 77.68%, 85.44% and 92.39%.

E REGRESSION WITH REJECTION EXPERIMENT DETAILS

This section presents the full results of the RwR experiment along with implementation details.

E.1 Data Description

For this experiment, we used eight regression datasets from the UCI Machine Learning Repository (Kelly et al., 2023). The datasets were accessed via the `uci_datasets` Python package, which provides preprocessed datasets with predefined train-test splits for benchmarking studies (`https://github.com/treforevans/uci_datasets`). A brief description of each dataset is provided below:

- **Concrete**: Predicts the compressive strength of concrete based on its ingredients and age. Contains 1030 instances with 8 features.
- **Wine**: Predicts wine quality based on physicochemical tests. The red wine subset contains 1599 instances with 11 features.
- **Airfoil**: Predicts the scaled sound pressure level of airfoil blade sections tested in an anechoic wind tunnel. Contains 1503 instances with 5 features.
- **Energy**: Predicts the energy efficiency of buildings based on building parameters. Contains 768 instances with 8 features.
- **Housing**: Predicts the median home value in Boston based on various housing features. Contains 506 instances with 13 features.
- **Solar**: Predicts the number of X-class solar flares in a 24-hour period. Contains 1066 instances with 10 features.
- **Forest**: Predicts the burned area of forest fires based on environmental factors. Contains 517 instances with 12 features.
- **Parkinsons**: Predicts UPDRS scores from individuals with early-stage Parkinson's disease. Contains 5875 instances with 20 features.

E.2 Implementation Details

In this experiment, we trained 16 regressor-calibrator pairs for each dataset. Each model in a pair was selected from one of the following options:

- **LR**: Standard linear regression.
- **RF**: Random forest without a maximum depth limit.
- **MLP**: A multilayer perceptron with a single hidden layer containing 64 units. This model uses ReLU activation and Adam optimizer. Training was performed with a batch size of 256, an initial learning rate of 0.0005, and a maximum of 800 iterations.
- **MLP2**: A multilayer perceptron with two hidden layers, each containing 64 units. Like the **MLP** model, it uses ReLU activation and the Adam optimizer, with the same training parameters (batch size of 256, initial learning rate of 0.0005, and a maximum of 800 iterations).

All models were implemented using the `scikit-learn` Python package (v1.2.1). Feature standardization was applied to all models – except for Random Forest – using `StandardScaler()` from `scikit-learn`, which removes the mean and scales features to unit variance. The models were trained using the squared loss objective function. To ensure reproducibility, we fixed the random seed at 42 for all algorithms.

We performed 10-fold cross-validation using splits provided by the `uci_datasets` package. In each fold, the dataset was partitioned into three subsets: 50% of the data were used to train the regressor $\hat{f}$, 40% to train the calibrator $\hat{R}$, and the remaining 10% were reserved for testing. The results reported for this experiment are the averages across the 10 folds.

E.3 Table from the Main Paper

Table 6: Comparative Performance of Learning with Rejection Methods, Measured by RwR Losses on the Testing Data.

	Concrete				Wine				Airfoil				Energy			
Method\Cost c	0.2	0.5	1.0	2.0	0.2	0.5	1.0	2.0	0.2	0.5	1.0	2.0	0.2	0.5	1.0	2.0
MLP+RF	**0.20**	**0.50**	**1.00**	**2.00**	**0.17**	**0.24**	**0.25**	**0.25**	**0.20**	0.50	0.99	1.91	0.20	0.46	0.80	1.32
RF+RF	**0.20**	**0.50**	**1.00**	**2.00**	0.18	0.28	0.30	0.31	**0.20**	0.50	**0.97**	**1.73**	**0.14**	**0.26**	**0.30**	**0.32**
NN+kNNRej	**0.20**	**0.50**	**1.00**	**2.00**	0.23	0.27	0.28	0.28	**0.20**	0.50	1.00	2.02	0.21	0.58	1.03	1.60
SelNet	0.75	1.09	2.82	3.16	0.18	**0.24**	**0.25**	**0.25**	0.21	**0.49**	0.99	1.84	0.26	0.49	0.86	1.39
	Housing				Solar				Forest				Parkinsons			
Method\Cost c	0.2	0.5	1.0	2.0	0.2	0.5	1.0	2.0	0.2	0.5	1.0	2.0	0.2	0.5	1.0	2.0
MLP+RF	**0.20**	**0.50**	**1.00**	**2.00**	0.23	**0.40**	0.54	**0.64**	**0.20**	0.51	1.08	**1.97**	0.20	0.50	1.00	2.00
RF+RF	**0.20**	**0.50**	1.01	2.07	0.23	0.41	0.57	0.73	**0.20**	0.51	**1.04**	1.98	**0.12**	**0.20**	**0.26**	**0.32**
NN+kNNRej	0.22	0.59	**1.00**	**2.00**	0.19	**0.40**	0.60	0.75	0.30	**0.50**	1.09	2.13	0.20	0.50	1.00	2.00
SelNet	1.27	1.31	2.23	2.47	**0.18**	0.45	**0.52**	0.68	0.54	0.78	1.25	2.17	0.20	0.60	1.17	1.97

E.4 Full Regression with Rejection Results

Table 7 presents the RwR losses for all 16 regressor-calibrator pairs, along with two benchmark RwR algorithms: SelectiveNet (Geifman and El-Yaniv, 2019b) and NN+kNNRej (Li et al., 2024). SelectiveNet is a deep learning approach with an integrated rejector, whereas NN+kNNRej is a no-rejection learning algorithm that achieved the best performance in the study by Li et al. (2024). We used the publicly available implementations of these algorithms from `https://github.com/hanzhao-wang/RwR`.

Table 7: Full RwR Experiment Results.

Data	Cost	LR + LR	LR + MLP	LR + MLP2	LR + RF	MLP + LR	MLP + MLP	MLP + MLP2	MLP + RF	MLP2 + LR	MLP2 + MLP	MLP2 + MLP2	MLP2 + RF	RF + LR	RF + MLP	RF + MLP2	RF + RF	SelNet	NN + kNNRej
Concrete	0.2	1.31	**0.20**	**0.20**	**0.20**	1.32	**0.20**	0.23	**0.20**	1.15	**0.20**	0.93	**0.20**	1.42	**0.20**	1.37	**0.20**	0.75	**0.20**
	0.5	1.61	**0.50**	**0.50**	**0.50**	1.61	**0.50**	0.53	**0.50**	1.45	**0.50**	1.24	**0.50**	1.71	**0.50**	1.72	**0.50**	1.09	**0.50**
	1	2.10	**1.00**	**1.00**	**1.00**	2.09	**1.00**	1.02	**1.00**	2.19	**1.00**	1.73	1.02	2.21	**1.00**	2.21	**1.00**	2.82	**1.00**
	2	3.08	**2.00**	**2.00**	**2.00**	3.08	**2.00**	2.02	**2.00**	3.24	**2.00**	2.81	2.01	3.87	2.03	3.27	**2.00**	3.16	**2.00**
Wine	0.2	0.22	0.21	0.21	0.19	0.19	0.19	0.18	**0.17**	0.20	0.20	0.19	**0.17**	0.21	0.21	0.20	0.18	0.18	0.23
	0.5	0.33	0.33	0.33	0.31	0.25	0.25	**0.24**	**0.24**	0.28	0.27	0.26	0.25	0.29	0.29	0.28	0.28	**0.24**	0.27
	1	0.36	0.35	0.37	0.34	**0.25**	**0.25**	**0.25**	**0.25**	0.27	0.27	0.28	0.27	0.30	0.29	0.30	0.30	**0.25**	0.28
	2	0.36	0.36	0.36	0.36	**0.25**	**0.25**	**0.25**	**0.25**	0.26	0.27	0.27	0.26	0.30	0.29	0.30	0.31	**0.25**	0.28
Airfoil	0.2	**0.20**	0.26	0.68	**0.20**	0.69	0.30	0.38	**0.20**	0.36	0.25	0.50	**0.20**	0.31	0.29	0.37	**0.20**	0.21	**0.20**
	0.5	0.50	0.56	0.97	0.50	0.98	0.60	0.73	0.50	0.70	0.56	0.82	0.50	0.62	0.59	0.68	0.50	**0.49**	0.50
	1	1.00	1.06	1.45	1.00	1.47	1.11	1.25	0.99	1.19	1.05	1.26	**0.96**	1.09	1.08	1.14	0.97	0.99	1.00
	2	2.06	2.08	2.41	1.94	2.43	2.11	2.21	1.91	2.27	1.91	2.15	1.84	1.98	1.99	1.99	**1.73**	1.84	2.02
Energy	0.2	0.77	0.52	0.31	0.26	0.44	0.43	0.27	0.20	0.23	0.23	0.27	0.20	0.17	0.16	0.17	**0.14**	0.26	0.21
	0.5	1.03	0.79	0.61	0.54	0.69	0.69	0.56	0.46	0.49	0.49	0.50	0.45	0.28	0.27	0.28	**0.26**	0.49	0.58
	1	1.46	1.20	1.03	0.85	1.06	1.18	0.98	0.80	0.79	0.77	0.78	0.71	0.31	0.32	0.32	**0.30**	0.86	1.03
	2	2.17	1.93	1.62	1.29	1.65	1.97	1.60	1.32	1.18	1.15	1.13	1.04	0.33	**0.32**	0.34	**0.32**	1.39	1.60
Housing	0.2	2.94	0.23	1.39	**0.20**	2.17	0.48	1.60	**0.20**	1.82	0.48	1.38	**0.20**	1.58	0.47	1.02	**0.20**	1.27	0.22
	0.5	3.19	0.53	1.70	**0.50**	2.41	0.84	1.84	**0.50**	2.13	0.78	1.65	**0.50**	1.84	0.75	1.33	**0.50**	1.31	0.59
	1	3.62	1.03	2.20	**1.00**	2.97	1.33	2.46	**1.00**	2.62	1.26	2.12	**1.00**	2.27	1.34	1.81	1.01	2.23	**1.00**
	2	4.64	2.44	3.27	2.04	3.95	2.33	3.28	**2.00**	3.49	2.38	3.19	2.01	3.18	2.31	2.73	2.07	2.47	**2.00**
Solar	0.2	0.19	0.23	0.35	0.24	0.20	0.23	0.30	0.23	0.20	0.25	0.38	0.24	0.19	0.24	0.33	0.23	**0.18**	0.19
	0.5	0.37	0.39	0.54	0.41	0.37	0.36	0.46	0.40	0.37	0.39	0.54	0.44	**0.35**	0.36	0.50	0.41	0.45	0.40
	1	**0.50**	0.53	0.62	0.54	0.51	0.54	0.64	0.54	0.52	0.57	0.63	0.55	0.55	0.59	0.70	0.57	0.52	0.60
	2	**0.60**	0.68	0.66	0.63	0.63	0.70	0.68	0.64	0.66	0.73	0.72	0.66	0.71	0.72	0.81	0.73	0.68	0.75
Forest	0.2	0.30	0.33	0.52	**0.20**	0.28	0.32	0.56	**0.20**	0.32	0.33	0.61	**0.20**	0.23	0.28	0.48	**0.20**	0.54	0.30
	0.5	0.60	0.65	0.83	0.51	0.58	0.65	0.90	0.51	0.63	0.67	0.87	0.51	0.54	0.59	0.82	0.51	0.78	**0.50**
	1	1.13	1.22	1.30	1.07	1.13	1.17	1.33	1.08	1.14	1.16	1.33	1.05	1.06	1.08	1.39	**1.04**	1.25	1.09
	2	2.03	1.96	1.91	2.00	2.00	1.95	**1.85**	1.97	2.03	1.98	1.89	1.97	1.95	1.96	2.06	1.98	2.17	2.13
Parkinsons	0.2	0.78	0.66	1.31	0.20	0.72	0.62	1.12	0.20	0.62	0.52	0.80	0.20	0.19	0.17	0.19	**0.12**	0.20	0.20
	0.5	1.08	0.96	1.60	0.50	1.02	0.94	1.41	0.50	0.91	0.83	1.07	0.50	0.33	0.28	0.29	**0.20**	0.60	0.50
	1	1.57	1.47	2.08	1.00	1.53	1.44	1.90	1.00	1.36	1.30	1.51	1.00	0.37	0.34	0.33	**0.26**	1.17	1.00
	2	2.57	2.48	3.02	2.00	2.58	2.52	2.89	2.00	2.26	2.15	2.33	1.89	0.35	0.37	0.33	**0.32**	1.97	2.00